

DELIVERY PERFORMANCE ANALYSIS AND STRATEGIC PLANNING IN LOGISTICS

Internship Report: Week 1

Project: Logistics Data Analysis and Predictive Modeling

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1. INTRODUCTION

Logistics is an important part of modern business operations because it ensures that products are transported from their source to the customer efficiently and reliably. The performance of a logistics operation directly affects delivery time, transportation costs, customer satisfaction, and overall business efficiency.

One of the major challenges faced by logistics organizations is delivery delay. Delays may occur because of several factors, including adverse weather conditions, long delivery distances, transportation conditions, vehicle characteristics, delivery modes, and operational constraints. Identifying these factors can help organizations take preventive measures and improve their delivery processes.

The increasing availability of logistics data provides an opportunity to use data analytics and machine learning for better decision-making. By analyzing historical delivery information, organizations can identify patterns, calculate performance indicators, determine factors associated with delays, and predict potential delivery risks.

This project focuses on the analysis of logistics delivery data using Python-based data science techniques. The project investigates delivery delays, weather conditions, delivery distance, delivery costs, vehicle types, customer ratings, and other operational factors. Exploratory Data Analysis is performed to understand the dataset and identify important patterns.

A Logistic Regression classification model is also developed to predict whether a delivery is likely to be delayed. The model achieved an accuracy of **89.56%** on the test dataset. The findings from the analysis are then converted into strategic recommendations for improving logistics performance.

2. BACKGROUND RESEARCH

Logistics performance depends on multiple internal and external factors. Understanding these factors is important for improving delivery reliability and controlling operational costs.

2.1 Weather Conditions

Weather conditions can significantly influence transportation operations. Rain, storms, fog, and other adverse conditions can reduce visibility, affect road conditions, increase travel time, and create safety challenges. Therefore, weather information can be useful when planning delivery schedules and contingency measures.

2.2 Delivery Distance

Delivery distance is an important factor in transportation planning. Longer delivery distances generally require more travel time, fuel, vehicle usage, and operational resources. Therefore, distance can have a significant effect on delivery cost and planning decisions.

2.3 Vehicle Type

Different vehicle types may have different capacities and operating characteristics. Vehicle selection can influence transportation efficiency, package handling, and delivery operations.

2.4 Delivery Mode

Different delivery modes may provide different levels of speed, flexibility, and operational efficiency. Analyzing delivery modes can help organizations understand whether specific modes are associated with better delivery performance.

2.5 Delivery Cost

Delivery cost represents the financial resources required to complete transportation activities. Understanding the relationship between delivery cost and factors such as distance can help organizations improve budgeting and cost management.

2.6 Customer Satisfaction

Customer satisfaction is an important indicator of logistics service quality. Customer ratings can provide feedback about the overall delivery experience and help organizations identify areas requiring improvement.

2.7 Role of Data Analytics in Logistics

Data analytics can help logistics organizations:

- Monitor delivery performance

- Identify delay patterns
- Analyze operational factors
- Control transportation costs
- Improve resource allocation
- Support route planning
- Predict potential delays
- Improve customer satisfaction

Therefore, combining exploratory data analysis with machine learning can provide useful insights for logistics decision-making.

3. PROJECT DEFINITION

3.1 Logistics Scenario

The project considers a logistics organization responsible for managing thousands of deliveries across multiple regions. The organization handles different types of packages using different delivery partners, vehicle types, and delivery modes.

The delivery operations are affected by several factors such as weather conditions, delivery distance, package weight, expected delivery time, transportation costs, and customer experience.

The organization is experiencing delivery delays and variations in transportation costs. Management requires a data-driven approach to understand the factors affecting delivery performance and identify potential delays before they occur.

The project therefore analyzes historical logistics data and develops a predictive model that can classify deliveries as delayed or not delayed. The resulting insights can support route planning, resource allocation, weather-based planning, cost management, and proactive delivery monitoring.

3.2 Problem Statement

Logistics organizations need to identify the major factors contributing to delivery delays and increased costs in order to improve operational efficiency and customer satisfaction.

3.3 Project Objectives

The main objectives of the project are:

1. To analyze logistics delivery data.
2. To understand the characteristics of the delivery dataset.
3. To identify patterns in delivery delays.
4. To calculate important logistics Key Performance Indicators.
5. To analyze the effect of weather conditions on delivery delays.
6. To analyze the relationship between delivery distance and delivery cost.
7. To compare delivery performance across different vehicle types.
8. To prepare the dataset for machine learning.
9. To develop a Logistic Regression model for predicting delivery delays.
10. To evaluate the performance of the predictive model.
11. To identify important findings from the analysis.
12. To provide strategic recommendations for improving logistics performance.

4. DATASET DESCRIPTION

The project uses a logistics delivery dataset containing **25,000 records and 15 features**. The dataset contains information related to delivery operations, transportation, environmental conditions, delivery performance, customer experience, and cost.

Dataset Summary

Attribute	Value
Number of Records	25,000
Number of Features	15
Target Variable	delay_target
Machine Learning Type	Binary Classification
Training Data	80%
Testing Data	20%

Main Data Categories

Category	Variables
Delivery Information	Delivery Partner, Package Type
Transportation	Vehicle Type, Delivery Mode
Location	Region
Environment	Weather Condition
Distance	Distance (km)
Package	Package Weight (kg)
Timing	Expected Time (hours)
Cost	Delivery Cost
Customer	Customer Rating
Outcome	Delivery Status / Delay Status

Target Variable

A binary target variable named `delay_target` was created for predictive modeling.

0 = Not Delayed

1 = Delayed

The target distribution was:

Delay Target	Number of Deliveries
0 — Not Delayed	18,331
1 — Delayed	6,669
Total	25,000

This means that the dataset contains approximately **73.32% non-delayed deliveries** and **26.68% delayed deliveries**.

Feature Selection

The following features were selected for the Logistic Regression model:

delivery_partner

package_type

vehicle_type

delivery_mode

region

weather_condition

distance_km

package_weight_kg

expected_time_hours

delivery_cost

```
print("\nDataset Information:")
print(df.info())

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45015 entries, 0 to 45014
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                     45015 non-null  int64
1   cb_person_cred_hist_length           45015 non-null  float64
2   loan_int_rate                        45013 non-null  float64
3   loan_intent                          45015 non-null  object
4   loan_percent_income                  45014 non-null  float64
5   person_education                     45015 non-null  object
6   person_emp_exp                       45014 non-null  float64
7   person_gender                        45015 non-null  object
8   person_home_ownership                45015 non-null  object
9   person_income                       45014 non-null  float64
10  previous_loan_defaults_on_file       45015 non-null  object
11  loan_status                          45015 non-null  float64
dtypes: float64(6), int64(1), object(5)
memory usage: 4.1+ MB
None
```


5. KEY PERFORMANCE INDICATORS

Key Performance Indicators (KPIs) provide measurable information about logistics performance. The following KPIs were calculated from the dataset.

KPI CALCULATION

KPI	Result
Delay Rate	26.68%
Successful Delivery Rate	73.32%
Average Delivery Cost	864.94
Average Customer Rating	3.67 / 5

5.1 Delay Rate

The delay rate was **26.68%**, meaning that approximately one out of every four deliveries experienced a delay.

This KPI is important because it provides a direct measure of delivery reliability.

5.2 Successful Delivery Rate

The successful delivery rate was **73.32%**.

This indicates that the majority of deliveries were completed successfully without being classified as delayed.

5.3 Average Delivery Cost

The average delivery cost was **864.94**.

This provides a baseline for evaluating transportation expenses and identifying opportunities for cost optimization.

5.4 Average Customer Rating

The average customer rating was **3.67 out of 5**.

This provides an indication of the overall customer experience associated with the delivery service.

6. DATA SCIENCE METHODOLOGIES

The project follows a structured data science workflow consisting of data understanding, preparation, exploratory analysis, predictive modeling, and strategic interpretation.

Project Workflow

Data Collection



Data Understanding



Data Cleaning



Exploratory Data Analysis



Feature Preparation



Model Training



Model Evaluation



Strategic Recommendations

6.1 Data Collection

The logistics dataset containing 25,000 delivery records was used as the primary source of analysis.

6.2 Data Understanding

The dataset was examined to understand:

- Number of rows
- Number of columns
- Data types
- Missing values
- Duplicate records
- Categorical variables
- Numerical variables
- Target variable

6.3 Data Cleaning

The dataset was checked and prepared for analysis. Data types and values were examined to ensure that the variables could be used for statistical analysis and machine learning.

6.4 Exploratory Data Analysis

EDA was performed to identify patterns and relationships between logistics variables.

The analysis included:

- Delivery status distribution
- Weather-based delay analysis
- Distance and delivery-cost relationship
- Vehicle-type delay comparison

6.5 Feature Preparation

The categorical and numerical features were separated.

Categorical features:

- Delivery partner
- Package type
- Vehicle type
- Delivery mode
- Region
- Weather condition

Numerical features:

- Distance
- Package weight
- Expected time
- Delivery cost

Categorical variables were converted into numerical representations using **One-Hot Encoding**.

6.6 Predictive Modeling

Logistic Regression was selected as the machine learning algorithm because the target variable contains two classes:

0 = Not Delayed

1 = Delayed

6.7 Model Evaluation

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Technologies Used

Technology	Purpose
Python	Programming and analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Visualization
Seaborn	Statistical visualization
Scikit-learn	Machine learning
One-Hot Encoding	Categorical feature encoding
Logistic Regression	Delay prediction

7. STRATEGIC ROADMAP

The project follows a six-phase strategic roadmap.

Phase 1 — Data Understanding

Understand the logistics dataset, identify important variables, and determine the characteristics of delivery operations.

Phase 2 — Performance Analysis

Calculate key performance indicators such as delay rate, successful delivery rate, average delivery cost, and customer rating.

Phase 3 — Risk Identification

Analyze operational and environmental factors that may be associated with delivery delays.

Phase 4 — Predictive Analytics

Develop a machine learning model to predict whether a delivery is likely to be delayed.

Phase 5 — Strategic Implementation

Use analytical findings to support:

- Route planning
- Resource allocation
- Weather-based planning
- Cost management
- Delivery monitoring

Phase 6 — Continuous Monitoring

Continuously monitor logistics KPIs and periodically retrain or improve the predictive model using new delivery data.

8. EXPLORATORY DATA ANALYSIS

8.1 Delivery Status Distribution

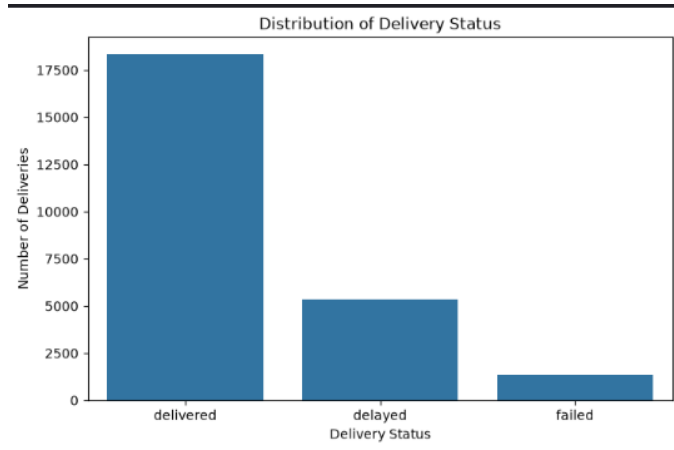


Figure 1: Delivery Status Distribution

Explanation

The delivery status distribution provides an overview of delivery outcomes within the dataset. The visualization helps establish the overall delivery-performance pattern and provides a foundation for further investigation of factors associated with delays.

8.2 Delay Rate by Weather Condition

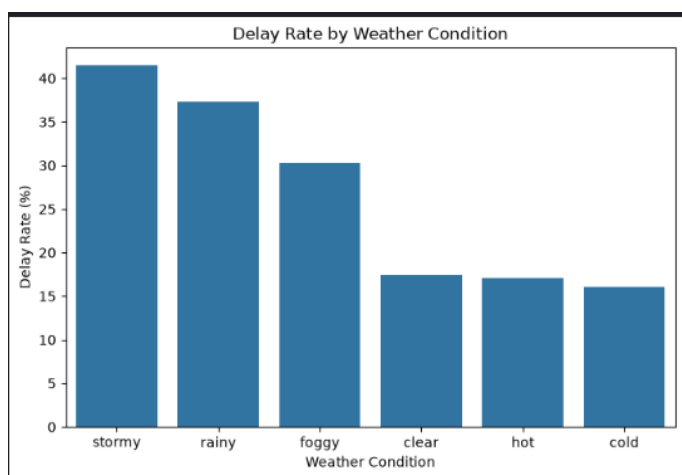


Figure 2: Delay Rate by Weather Condition

Weather Condition	Delay Rate
Stormy	41.45%
Rainy	37.35%
Foggy	30.32%
Clear	17.43%
Hot	17.12%
Cold	16.02%

Explanation

Weather conditions showed a noticeable difference in delivery delay rates. Stormy weather recorded the highest delay rate at 41.45%, followed by rainy weather at 37.35% and foggy weather at 30.32%.

In comparison, clear, hot, and cold conditions recorded lower delay rates of 17.43%, 17.12%, and 16.02%, respectively.

The results indicate that adverse weather conditions are an important operational consideration when planning delivery schedules and contingency measures.

8.3 Distance vs Delivery Cost

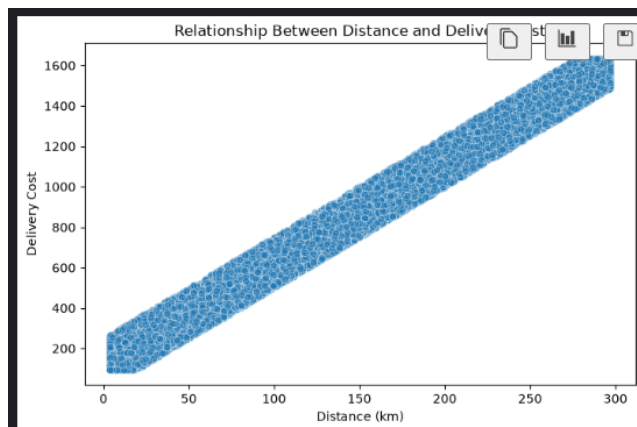


Figure 3: Relationship Between Distance and Delivery Cost

Explanation

The analysis identified a very strong positive relationship between delivery distance and delivery cost. The calculated correlation coefficient was approximately 0.99.

This indicates that deliveries involving greater distances tend to have higher delivery costs. Therefore, delivery distance should be considered during route planning, transportation budgeting, and cost-control activities.

Although the correlation is very strong, correlation represents an association between variables and should not by itself be interpreted as proof of causation.

8.4 Delay Rate by Vehicle Type

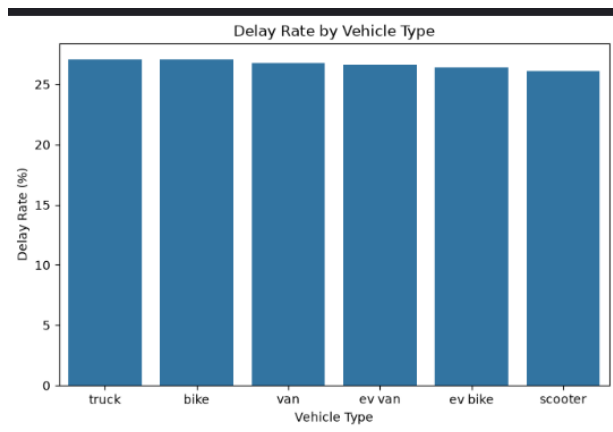


Figure 4: Delay Rate by Vehicle Type

Explanation

The analysis shows that delay rates across vehicle types were relatively similar. The observed rates ranged from approximately 26.14% to 27.04%.

The relatively small difference suggests that vehicle type alone does not create a major difference in delivery-delay risk within this dataset. Other factors, including weather conditions, distance, delivery mode, and operational conditions, may therefore need to be considered together when evaluating delivery risk.

9. PREDICTIVE MODELING

9.1 Model Selection

Logistic Regression was selected as the predictive machine learning algorithm for this project. Logistic Regression is a classification algorithm suitable for predicting binary outcomes.

In this project, the model predicts whether an individual delivery is delayed.

Target Variable

0 = Not Delayed

1 = Delayed

9.2 Model Features

The following features were used:

- Delivery Partner
- Package Type
- Vehicle Type
- Delivery Mode
- Region
- Weather Condition
- Distance
- Package Weight
- Expected Time
- Delivery Cost

9.3 Data Preprocessing

Before training the model, categorical and numerical variables were separated. Categorical variables were converted into numerical representations using One-Hot Encoding with unknown categories handled safely. Numerical variables were passed directly through the preprocessing pipeline.

The dataset was divided into 80% training data and 20% testing data. Stratified sampling was used to maintain the distribution of the target classes in both datasets.

Dataset	Percentage	Records
Training Set	80%	20,000
Testing Set	20%	5,000
Total	100%	25,000

9.4 Model Performance

Main result

LOGISTIC REGRESSION ACCURACY: 89.56%

Explanation

The Logistic Regression model achieved an overall accuracy of 89.56% on the testing dataset.

This indicates that the model correctly classified approximately 89.56% of the deliveries in the test set as either delayed or not delayed.

The result demonstrates that the selected logistics features contain useful information for predicting delivery delays.

9.5 Confusion Matrix

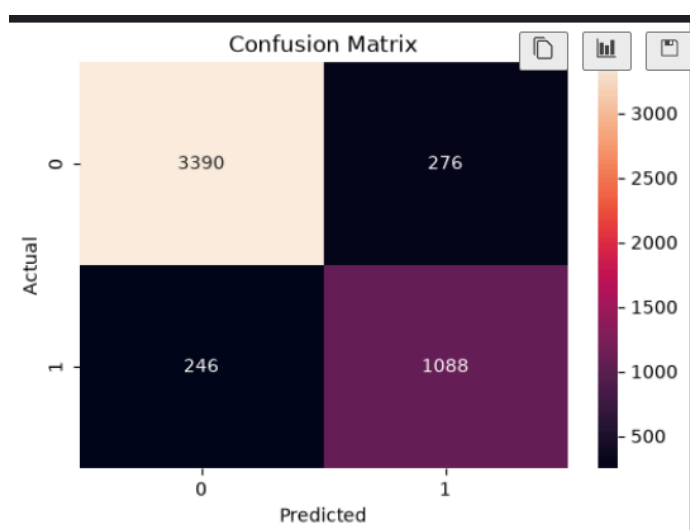


Figure 5: Confusion Matrix of Logistic Regression Model

Explanation

The confusion matrix compares the actual delivery-delay classes with the classes predicted by the Logistic Regression model. It shows the number of correctly and incorrectly classified deliveries.

Correct predictions indicate deliveries that were correctly classified as delayed or not delayed, while incorrect predictions represent cases where the model predicted the wrong class.

The confusion matrix provides more detailed information than accuracy alone because it shows how the model performs separately for each class.

9.6 Classification Performance

Metric	Not Delayed	Delayed
Precision	0.93	0.80
Recall	0.92	0.82
F1-score	0.93	0.81

The model performed particularly well in identifying non-delayed deliveries. For the delayed class, the precision was 0.80, recall was 0.82, and F1-score was 0.81. This indicates that the model provides useful delay-risk predictions, although some delayed deliveries may still be incorrectly classified.

10. KEY FINDINGS

The major findings from the project are summarized below.

Finding 1 — Delivery Delays

Approximately 26.68% of deliveries were delayed.

This indicates that delivery delays represent a significant operational consideration.

Finding 2 — Successful Deliveries

The successful delivery rate was 73.32%.

The majority of deliveries were completed successfully, but there remains an opportunity to reduce the proportion of delayed deliveries.

Finding 3 — Delivery Cost

The average delivery cost was 864.94.

This provides a baseline for future transportation cost monitoring and optimization.

Finding 4 — Customer Rating

The average customer rating was 3.67/5.

This indicates a moderate overall customer experience and provides a KPI for monitoring service quality.

Finding 5 — Weather Impact

Stormy weather recorded the highest delay rate at 41.45%.

Rainy and foggy weather also recorded relatively high delay rates.

Finding 6 — Distance and Cost

Distance and delivery cost had a correlation of approximately 0.99, indicating a very strong positive relationship.

Finding 7 — Vehicle Type

Vehicle-type delay rates were relatively similar, ranging from approximately 26.14% to 27.04%.

Finding 8 — Predictive Model

The Logistic Regression model achieved 89.56% accuracy.

Finding 9 — Overall Insight

The analysis indicates that logistics performance should not be evaluated using a single factor. Weather, distance, cost, delivery characteristics, and other operational variables should be considered together when making logistics decisions.

11. STRATEGIC RECOMMENDATIONS

11.1 Weather-Based Contingency Planning

Since stormy and rainy conditions showed substantially higher delay rates, logistics organizations should introduce weather-based contingency plans. Additional delivery time buffers, alternative routes, and flexible scheduling can be considered during severe weather.

11.2 Predictive Delay Monitoring

The Logistic Regression model can be used as a preliminary risk-screening tool to identify deliveries that have a higher probability of being delayed. High-risk deliveries can receive additional operational attention.

11.3 Distance-Based Cost Management

Since distance has a very strong positive relationship with delivery cost, distance should be considered during route planning and transportation budgeting. Reducing unnecessary travel distance can help control transportation expenditure.

11.4 Improved Resource Allocation

Vehicles and delivery resources should be allocated based on operational requirements, expected demand, distance, weather conditions, and delivery risk.

11.5 Regular KPI Monitoring

Logistics management should regularly monitor delay rate, successful delivery rate, average delivery cost, and customer rating. Continuous KPI monitoring can help identify performance changes at an early stage.

11.6 Real-Time Data Integration

Future implementations can integrate real-time weather, traffic, GPS, and delivery-status information. This would allow the predictive system to respond to changing operational conditions.

11.7 Route Optimization

Advanced route optimization techniques can be explored to reduce travel distance, delivery time, fuel consumption, and transportation costs.

11.8 Model Improvement

Future versions of the project can compare Logistic Regression with algorithms such as Decision Trees, Random Forest, Gradient Boosting, and other classification techniques to determine whether predictive performance can be improved.

12. EXPECTED OUTCOMES

The implementation of the proposed analytical and strategic approach is expected to provide several operational benefits.

12.1 Reduced Delivery Delays

Predictive identification of high-risk deliveries can allow logistics teams to take preventive action before delays occur.

12.2 Better Route Planning

Distance and operational information can support more efficient route planning.

12.3 Improved Resource Utilization

Data-driven allocation of vehicles and delivery resources can improve operational efficiency.

12.4 Better Cost Control

Understanding the relationship between distance and delivery cost can support more effective transportation budgeting.

12.5 Improved Customer Satisfaction

Reducing delivery delays and improving delivery reliability can contribute to a better customer experience.

12.6 Early Risk Identification

The predictive model can help identify potentially delayed deliveries before completion.

12.7 Data-Driven Decision Making

The overall project provides a framework for making logistics decisions based on actual delivery data rather than relying only on assumptions.

13. CONCLUSION

This project demonstrates how data analysis and machine learning can be applied to improve logistics decision-making. The analysis examined 25,000 delivery records and investigated important factors including weather conditions, delivery distance, vehicle type, delivery cost, customer ratings, and delivery outcomes.

The analysis found that 26.68% of deliveries were delayed, while the successful delivery rate was 73.32%. Stormy weather recorded the highest delay rate at 41.45%, while rainy and foggy conditions also showed relatively high delay rates. The analysis also identified a very strong positive relationship between delivery distance and delivery cost, with a correlation coefficient of approximately 0.99.

A Logistic Regression model was developed to predict delivery delays using both categorical and numerical features. The model achieved an accuracy of 89.56% on the test dataset. The result demonstrates the potential of machine learning as a supporting tool for identifying delivery-delay risk.

Based on the findings, recommendations were proposed for weather-based contingency planning, predictive delay monitoring, distance-based cost management, resource allocation, KPI monitoring, real-time data integration, and route optimization.

Overall, the project provides a practical data-driven framework for understanding logistics performance and supporting proactive operational planning. Future work can focus on integrating real-time traffic and weather data, comparing multiple machine learning algorithms, implementing advanced route optimization, and continuously updating the predictive model with new delivery data.

14. REFERENCES

References

1. Kaggle. *Logistics Delivery Dataset*.

